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COURSE: Autonomous Systems Lab 1

GROUP: 4

TRUCK PLATOONING SCENARIO: Platoon Splitting

Scenario Context

The truck platoon of three follower trucks and one lead truck approaches a point of divergence where the trucks would split into two sub-platoons of two trucks each, then take different routes. The lead truck's driver initiates manual override of the platoon V2V communication. The lead truck maintains its status and leads one of the sub-platoons while one out of the three follower trucks assumes the leadership status of the second sub-platoon. V2V communication is reestablished between the trucks in each platoon and gap reconfiguration is synchronized.

The platoon must:

1. Detect upcoming divergence
2. Coordinate separation
3. Create safe gaps
4. Split control groups
5. Stabilize resulting sub-platoons

Timing Phases

PHASE	DESCRIPTION	TYPICAL TIMING CHARACTERISTICS
1	Route divergence detection	Seconds to minutes before split
2	Split negotiation and coordination	100ms - several seconds
3	Gap creation	2-10 seconds
4	Physical separation	1-5 seconds
5	Topology reconfiguration	<1s – a few seconds
6	Stabilization of sub-platoons	5 – 20 seconds

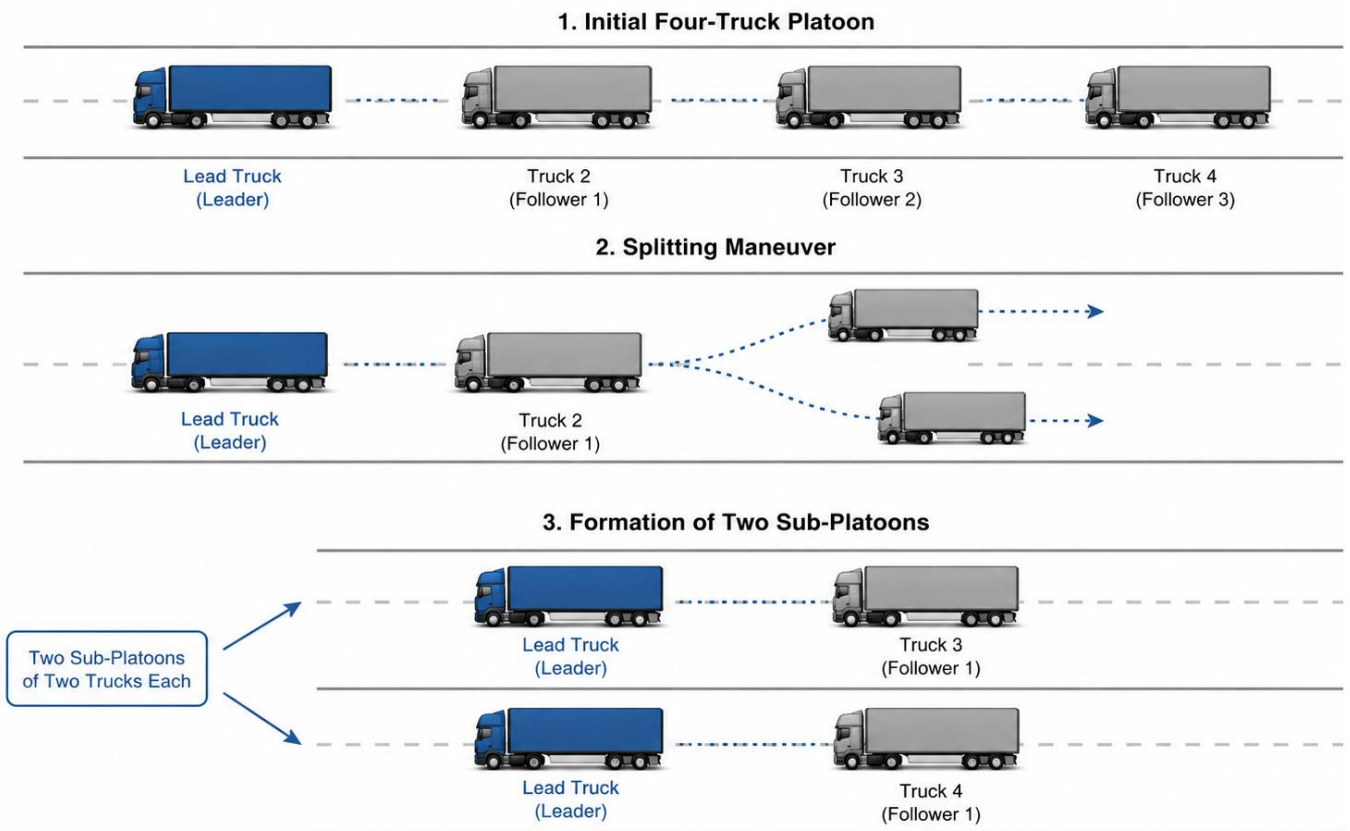


Fig.1 Truck Platoon Splitting Illustration (this image was generated by Chatgpt using my prompt)